

Authored Grade-9 Force and Laws of Motion spine covering balanced/unbalanced for: Source questions

IIT-JEE · 2007 · Paper I · Q3

Official past paper, faithful restatement of the original

Two equal masses m lie on a smooth horizontal surface and are joined by a light string of total length $2a$. The midpoint of the initially taut string is pulled vertically upward by a small constant force F . When the masses are separated by $2x$, determine the magnitude of the acceleration of either mass. (a) (A) $(F/2m) \cdot a/\sqrt{a^2-x^2}$ (b) (B) $(F/2m) \cdot x/\sqrt{a^2-x^2}$ (c) (C) $(F/2m) \cdot x/a$ (d) (D) $(F/2m) \cdot \sqrt{a^2-x^2}/x$

Conditions

- Horizontal surface is frictionless.
- String is light and remains taut.
- Each half-string has length a .

Answer and working

Option (B): $a_{\text{mass}} = F x / [2m \sqrt{a^2-x^2}]$.

1. Let the midpoint rise by y . String geometry gives $x^2 + y^2 = a^2$, so $y = \sqrt{a^2-x^2}$.
2. At the massless midpoint, vertical force balance gives $2T(y/a)=F$, hence $T=Fa/(2y)$.

3. The horizontal force on either mass is $T(x/a)$, so its acceleration is $F x/[2m \sqrt{a^2-x^2}]$.

IIT-JEE · 2007 · Paper I · Q10

Official past paper, faithful restatement of the original

A block moving on a rough horizontal surface stops after losing its kinetic energy to friction. The same surface is then inclined at 30 degrees and the block is projected up the incline with the same initial speed. Judge these claims: (1) the energy dissipated by friction before stopping is smaller on the incline; (2) the coefficient of friction becomes smaller when the plane is inclined. (a) (A) Both true and (2) explains (1) (b) (B) Both true but (2) does not explain (1) (c) (C) (1) true, (2) false (d) (D) (1) false, (2) true

Conditions

- Same pair of surfaces is used.
- Initial speed is the same in both trials.
- Incline angle is 30 degrees.

Answer and working

Option (C): statement (1) is true and statement (2) is false.

1. On the horizontal surface, the entire initial kinetic energy is dissipated by friction before the block stops.
2. Going up the incline, part of the initial kinetic energy becomes gravitational potential energy, so friction dissipates less than the full initial kinetic energy.
3. The coefficient of friction is a property of the contacting surfaces in this model; tilting changes the normal force, not the coefficient.

IIT-JEE · 2008 · Paper II · Q33

Official past paper, faithful restatement of the original

Judge the pair of statements: (1) pulling a heavy body across level ground is easier than pushing it when the applied force is oblique; (2) frictional force depends on the nature of the surfaces in contact. (a) (A) Both true and (2) explains (1) (b) (B) Both true but (2) does not explain (1) (c) (C) (1) true, (2) false (d) (D) (1) false, (2) true

Conditions

- Compare oblique pull and oblique push on the same level surface.

Answer and working

Option (B): both statements are true, but statement (2) is not the explanation of statement (1).

1. An upward component while pulling reduces the normal reaction; a downward component while pushing increases it.
2. Because limiting/kinetic friction scales with the normal reaction, pulling requires less horizontal effort under otherwise comparable conditions.

3. Although friction also depends on surface nature through the coefficient, that fact does not explain the pull-versus-push difference for the same surfaces.

IIT-JEE · 2011 · Paper I · Q41

Official past paper, faithful restatement of the original

A block is on a plane inclined at 45 degrees. The coefficient of friction is μ . The force needed to just push the block upward is three times the force needed to just prevent it from sliding downward. If $N = 10 \mu$, find N .

Conditions

- Both forces act parallel to the incline.
- Each case is at limiting equilibrium.

Answer and working

5.

1. For impending upward motion, friction acts down the plane: $P_{\text{up}} = mg(\sin 45^\circ + \mu \cos 45^\circ)$.
2. For impending downward slip, friction acts up the plane: $P_{\text{hold}} = mg(\sin 45^\circ - \mu \cos 45^\circ)$.

3. Using $P_{\text{up}} = 3P_{\text{hold}}$ gives $1+\mu=3(1-\mu)$, so $\mu=1/2$ and $N=5$.

IIT-JEE · 2011 · Paper II · Q34

Official past paper, faithful restatement of the original

A 0.18 kg block on a rough horizontal surface is attached to a spring of force constant 2 N/m. The spring is initially unstretched. The block is given a horizontal impulse and first comes to rest after moving 0.06 m. With coefficient of friction 0.10 and $g = 10 \text{ m/s}^2$, its initial speed is $N/10 \text{ m/s}$. Find N .

Conditions

- Spring is initially unstretched.
- Kinetic friction coefficient is 0.10.
- Use $g=10 \text{ m/s}^2$.

Answer and working

4.

1. At the first stopping point, the initial kinetic energy has become spring energy plus work against friction.
2. Write $(1/2)mv^2 = (1/2)kx^2 + \mu m g x$ with $m=0.18$, $k=2$, $x=0.06$ and $\mu=0.10$.

3. The right side is $0.0036 + 0.0108 = 0.0144$ J, so $0.09v^2 = 0.0144$ and $v = 0.4$ m/s; hence $N = 4$.

JEE (Advanced) · 2014 · Paper 2 · Q19

Official past paper, faithful restatement of the original

Two touching blocks $m_1=1$ kg and $m_2=2$ kg rest on an incline of angle θ . The plane is smooth under m_1 ; for m_2 , static and kinetic friction coefficients are both 0.3. Determine which friction expression applies to m_2 at $\theta = 5^\circ, 10^\circ, 15^\circ$ and 20° . Candidate expressions are: (1) $m_2 g \sin \theta$; (2) $(m_1+m_2)g \sin \theta$; (3) $0.3 m_2 g \cos \theta$; (4) $0.3(m_1+m_2)g \cos \theta$. (a) (A) P-1, Q-1, R-1, S-3 (b) (B) P-2, Q-2, R-2, S-3 (c) (C) P-2, Q-2, R-2, S-4 (d) (D) P-2, Q-2, R-3, S-3

Conditions

- $m_1=1$ kg; $m_2=2$ kg.
- Only m_2 has friction with the plane, $\mu=0.3$.
- Use $\tan 5.5^\circ \approx 0.1$, $\tan 11.5^\circ \approx 0.2$, $\tan 16.5^\circ \approx 0.3$.

Answer and working

Option (D): $5^\circ \rightarrow 2$, $10^\circ \rightarrow 2$, $15^\circ \rightarrow 3$, $20^\circ \rightarrow 3$.

1. If both blocks can remain at rest together, friction on m_2 must balance the total downslope weight component: $f_{\text{req}} = (m_1 + m_2)g \sin \theta$.
2. The maximum available friction at m_2 is $f_{\text{max}} = \mu m_2 g \cos \theta$. Static balance is possible only while $(m_1 + m_2)\tan \theta \leq \mu m_2$, i.e. $\tan \theta \leq 0.2$.
3. Thus 5° and 10° use expression 2; above the threshold the system slides and friction is $\mu m_2 g \cos \theta$, expression 3, for 15° and 20° .

JEE (Advanced) · 2020 · Paper 1 · Q13

Official past paper, faithful restatement of the original

A uniform metre scale is supported horizontally on two index fingers initially at 0.00 cm and 90.00 cm. Both fingers are moved slowly toward the centre. The slipping finger alternates because static and kinetic friction differ. With $\mu_s=0.40$ and $\mu_k=0.32$, find the final distance x_R of the right finger from the 50.00 cm centre at the instant it stops and the left finger starts slipping again.

Conditions

- Scale is uniform.
- Finger motion is quasistatic.
- $\mu_s=0.40$; $\mu_k=0.32$.

Answer and working

25.6 cm.

1. While the left finger slips and the right sticks at 90 cm, limiting transition occurs when $\mu_k N_L = \mu_s N_R$. Static torque balance gives $N_L/N_R=40/(50-l)$, yielding $l=18$ cm.

2. Now the left finger sticks at 18 cm while the right slips inward. For supports at 18 cm and r , torque balance gives $N_R/N_L = 32/(r-50)$.
3. At the next transition $\mu_k N_R = \mu_s N_L$, so $0.32 \cdot 32/(r-50) = 0.40$. Hence $r-50 = 25.6$ cm.

JEE (Advanced) · 2014 · Paper 1 · Q8

Official past paper, faithful restatement of the original

A uniform ladder of mass m leans against a wall at angle θ and is on the verge of slipping. Let the wall and floor friction coefficients be μ_1 and μ_2 , with normal reactions N_1 at the wall and N_2 at the floor.

Identify the source-correct limiting-equilibrium statements. (a) (A) $\mu_1=0$, $\mu_2 \neq 0$ and $N_2 \tan(\theta) = mg/2$ (b) (B) $\mu_1 \neq 0$, $\mu_2=0$ and $N_1 \tan(\theta) = mg/2$ (c) (C) $\mu_1 \neq 0$, $\mu_2 \neq 0$ and $N_2 = mg/(1 + \mu_1 \mu_2)$ (d) (D) $\mu_1=0$, $\mu_2 \neq 0$ and $N_1 \tan(\theta) = mg/2$

Conditions

- The ladder is in static limiting equilibrium.
- Friction acts at the impending-slip limits where present.

Answer and working

Options (C) and (D).

1. With friction at both contacts, horizontal balance gives $N_1 = \mu_2 N_2$ and vertical balance gives $N_2 + \mu_1 N_1 = mg$.

2. Eliminating N_1 yields $N_2 = mg/(1 + \mu_1 \mu_2)$, establishing option (C).
3. If the wall is smooth ($\mu_1 = 0$), taking moments about the foot gives $N_1 L \sin(\theta) = mg(L/2)\cos(\theta)$, hence $N_1 \tan(\theta) = mg/2$, establishing option (D).

NEET (UG) · 2020 · English Set E1 · Q160

Official past paper, faithful restatement of the original

Two masses, 4 kg and 6 kg, are connected by a massless string over a frictionless pulley. Find the magnitude of the system acceleration in terms of g . (a) (1) g (b) (2) $g/2$ (c) (3) $g/5$ (d) (4) $g/10$

Conditions

- String is massless.
- Pulley is frictionless.

Answer and working

Option (3): $g/5$.

1. For an ideal Atwood machine, the net driving force is $(6-4)g=2g$.
2. The total accelerated mass is 10 kg, so $a=2g/10=g/5$.

IIT-JEE · 2009 · Paper I · Q49

Official past paper, faithful restatement of the original

In an inertial frame, the resultant of all external forces on a system of particles is zero. Which quantity is guaranteed not to change with time? (a) (A) Linear momentum of the system (b) (B) Kinetic energy of the system (c) (C) Angular momentum of the system (d) (D) Potential energy of the system

Conditions

- The observation is made from an inertial frame.

Answer and working

Option (A) only: total linear momentum is constant.

1. For a system of particles, dP/dt equals the resultant external force.
2. Since the resultant external force is zero, $dP/dt=0$ and total linear momentum is constant.

3. Zero net force does not by itself guarantee zero external torque or conservation of kinetic or potential energy.

IIT-JEE · 2009 · Paper II · Q40

Official past paper, faithful restatement of the original

A bead can slide without friction on a wire shaped as the parabola $y=kx^2$, with y vertical. The wire is accelerated horizontally with constant acceleration a . Find the distance from the y -axis of the new equilibrium position where the bead can remain at rest relative to the wire. (a) (A) $a/(gk)$ (b) (B) $a/(2gk)$ (c) (C) $2a/(gk)$ (d) (D) $a/(4gk)$

Conditions

- The bead is frictionless on the wire.
- The required equilibrium is relative to the accelerating wire.

Answer and working

Option (B): $a/(2gk)$.

1. In the accelerating wire frame, the bead experiences gravity downward and a horizontal pseudo-force ma opposite the wire acceleration.

2. The effective potential along the wire is $U(x) = mgkx^2 \pm max$, with the sign depending on the acceleration direction.
3. Equilibrium requires $dU/dx=0$, so $2mgkx \pm ma=0$ and the distance magnitude is $|x|=a/(2gk)$.

IIT-JEE · 2009 · Paper II · Q55

Official past paper, faithful restatement of the original

A light inextensible string passes over a smooth fixed pulley and connects masses 0.36 kg and 0.72 kg. The system is released from rest. Taking $g=10 \text{ m/s}^2$, find the work in joules done by the string on the 0.36 kg block during the first second.

Conditions

- String is light and inextensible.
- Pulley is smooth.
- The system starts from rest.
- Use $g=10 \text{ m/s}^2$.

Answer and working

8 J.

1. The common acceleration is $a=(0.72-0.36)g/(0.72+0.36)=10/3 \text{ m/s}^2$.
2. For the 0.36 kg block moving upward, $T-mg=ma$, so $T=0.36(10+10/3)=4.8 \text{ N}$.

3. Its displacement in the first second is $s = \frac{1}{2}at^2 = \frac{5}{3} \text{ m}$, and string tension acts along the displacement, so $W = Ts = 4.8 \times \frac{5}{3} = 8 \text{ J}$.

JEE Main · 2026 · Session 2 · 06 Apr 2026 · Shift 2 · Q46

Official past paper, faithful restatement of the original

A block takes time t to slide from rest down a 45° inclined plane. When the same plane is made frictionless, the time becomes $t/2$. If the original coefficient of friction is $\alpha/100$, find α .

Conditions

- The block starts from rest in both cases and traverses the same distance along the plane.

Answer and working

75.

1. For the rough 45° plane, $a_r = g(\sin 45^\circ - \mu \cos 45^\circ) = g(1 - \mu)/\sqrt{2}$. For the smooth plane, $a_s = g/\sqrt{2}$.
2. For the same distance from rest, time is proportional to $1/\sqrt{a}$. Since the smooth time is $t/2$, $a_s = 4a_r$.

3. Therefore $1=4(1-\mu)$, giving $\mu=3/4=75/100$ and $\alpha=75$.

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